

UNIT I: MATRICES, RANK, SYSTEM OF EQUATIONS & EIGENVALUE THEOREMS

Weightage: 25% | Syllabus Section 1

1. Rank, Echelon Form & Normal Form

Rank $\rho(A)$: Number of non-zero rows in row-echelon form. Invariant under elementary row/column transformations.

Normal Form (Canonical Form): Using both row & column operations, reduce A to $[I_r \ 0; \ 0 \ 0]$. Non-singular

matrices P and Q exist such that $PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$, where $\rho(A) = r$.

Gauss-Jordan Inverse: Augment $[A \mid I]$. Apply row operations until $[I \mid A^{-1}]$ is obtained.

2. Consistency of Linear Systems ($AX = B$)

Rouché-Capelli Theorem: Form augmented matrix $[A \mid B]$.

Rank Relation	Consistency	Nature of Solution
$\rho(A) \neq \rho([A \mid B])$	Inconsistent	No Solution
$\rho(A) = \rho([A \mid B]) = n$	Consistent	Unique Solution
$\rho(A) = \rho([A \mid B]) = r < n$	Consistent	∞ Solutions ($n-r$ free params)

Homogeneous ($AX = 0$): Always consistent (trivial $X = 0$). Non-trivial (infinite) solution exists if and only if $|A| = 0$ ($\rho(A) < n$).

3. Eigenvalues & Cayley-Hamilton Theorem

Characteristic Equation: $|A - \lambda I| = 0 \implies \lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$.

- $S_1 = \text{Tr}(A) = \sum \lambda_i$
- $S_2 = M_{11} + M_{22} + M_{33}$
- $|A| = \prod \lambda_i$

Properties: Eigenvalues of A^{-1} are $1/\lambda_i$; of A^k are λ_i^k .

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation: $A^3 - S_1A^2 + S_2A - |A|I = 0$.

Application: Multiply by $A^{-1} = \frac{1}{|A|}[A^2 - S_1A + S_2I]$.

UNIT II: DIFFERENTIAL & INTEGRAL CALCULUS, CURVATURE & BETA-GAMMA FUNCTIONS

Weightage: 25% | Syllabus Section 2

1. Rolle's & Mean Value Theorems

Rolle's Theorem: If $f(x)$ is (1) continuous on $[a, b]$, (2) differentiable on (a, b) , (3) $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$.

Lagrange's MVT: Conditions (1) & (2) hold $\implies \exists c \in (a, b)$ such that:

$f'(c) = \frac{f(b)-f(a)}{b-a}$.

Cauchy's MVT: For $f(x), g(x)$:

$\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$.

Taylor's Series about a:

$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$

Maclaurin's Series: Set $a = 0$.

2. Curvature & Radius of Curvature (ρ)

Curvature κ : Rate of turning of tangent: $\kappa = \frac{d\psi}{ds} = \frac{1}{\rho}$.

Form	Formula for Radius of Curvature ρ
Cartesian $y = f(x)$	$\rho = \frac{[1+(y')^2]^{3/2}}{ y'' }$
Parametric $x(t), y(t)$	$\rho = \frac{(x'^2+y'^2)^{3/2}}{ x'y''-y'x'' }$
Polar $r = f(\theta)$	$\rho = \frac{(r^2+r'^2)^{3/2}}{ r^2+2r_1'^2-rr_2 }$
Pedal $p = f(r)$	$\rho = r \frac{dr}{dp}$

Centre of Curvature:

$\bar{x} = x - \frac{y_1(1+y_1^2)}{2}$; $\bar{y} = y + \frac{(1+y_1^2)}{2}$

$/ y_2$.

3. L'Hospital & Beta-Gamma Functions

L'Hospital's Rule: For $0/0$ or ∞/∞ :

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$.

Gamma Function: $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$ ($n > 0$).

• $\Gamma(n+1) = n\Gamma(n) = n!$ (for integer n).

• $\Gamma(\frac{1}{2}) = \sqrt{\pi}$; $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin(n\pi)}$.

Beta Function: $B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

Fundamental Relation:

$B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

Duplication formula: $\Gamma(m)\Gamma(m+\frac{1}{2}) =$

$\frac{\sqrt{\pi}}{2^{2m-1}}\Gamma(2m)$.

UNIT III: MULTIVARIABLE CALCULUS, EXTREMA & VECTOR DIFFERENTIAL CALCULUS

Weightage: 25% | Syllabus Section 3

1. Partial Derivatives, Euler's Theorem & Extrema

Euler's Theorem on Homogeneous Functions: If $u(x, y)$ is homogeneous of degree n :

- $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu.$
- $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u.$

Total Derivative: $du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy.$

Discriminant: $\Delta = rt - s^2$	Value of $r = f_{xx}$	Conclusion for Point (a, b)
$rt - s^2 > 0$	$r < 0$	Local Maximum
$rt - s^2 > 0$	$r > 0$	Local Minimum
$rt - s^2 < 0$	Any sign	Saddle Point (Neither max nor min)
$rt - s^2 = 0$	Any	Inconclusive (Further tests required)

Lagrange's Multipliers: Optimize $f(x, y, z)$ subject to $\phi(x, y, z) = 0$. Auxiliary function:
 $\nabla(f + \lambda\phi) = 0$.

2. Vector Differential Calculus: $\nabla\phi, \nabla\cdot\mathbf{F}, \nabla\times\mathbf{F}$

Del Operator: $\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}.$

Operation	Formula	Physical Meaning & Condition
Gradient ($\nabla\phi$)	$\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$	Normal vector to surface $\phi = c$. Max rate of increase = $ \nabla\phi $.
Directional Derivative	$D_a\phi = \text{abla}\phi \cdot \frac{\mathbf{a}}{ \mathbf{a} }$	Rate of change of ϕ along direction vector $\mathbf{a}$.
Divergence ($\nabla\cdot\mathbf{F}$)	$\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$	Net outward flux per volume. $\nabla \cdot \vec{F} = 0 \implies$ Solenoidal Field .
Curl ($\nabla\times\mathbf{F}$)	Determinant with $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$	Rotational vorticity. $\nabla \times \vec{F} = 0 \implies$ Irrotational / Conservative Field ($F = \text{abla}\phi$).

Fundamental Vector Null Identities:

- $\text{abla} \times (\text{abla}\phi) = 0$ (Curl of any gradient is identically zero).
- $\text{abla} \cdot (\text{abla} \times \mathbf{F}) = 0$ (Divergence of any curl is identically zero).

UNIT IV: CONVERGENCE TESTS OF SEQUENCES & SERIES AND FOURIER EXPANSIONS

Weightage: 25% | Syllabus Section 4

1. Infinite Series Convergence Tests

Test Name	Test Limit Ratio: L	Convergence Criterion
D'Alembert Ratio Test	$\lim_{n \rightarrow \infty} \left \frac{u_n}{u_{n+1}} \right = L$	$L > 1 \implies \text{Conv}; L < 1 \implies \text{Div}; L = 1 \implies \text{Fails}$
Raabe's Test (Use when Ratio fails)	$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = L$	$L > 1 \implies \text{Conv}; L < 1 \implies \text{Div}; L = 1 \implies \text{Fails}$
Logarithmic Test	$\lim_{n \rightarrow \infty} n \ln \left(\frac{u_n}{u_{n+1}} \right) = L$	$L > 1 \implies \text{Conv}; L < 1 \implies \text{Div}$
Cauchy's Root Test	$\lim_{n \rightarrow \infty} (u_n)^{1/n} = L$	$L < 1 \implies \text{Conv}; L > 1 \implies \text{Div}$
Leibnitz Rule (Alternating)	$\sum (-1)^{n-1} u_n$	Converges if (1) $u_{n+1} \leq u_n$, (2) $\lim u_n = 0$

- Absolute Convergence:** $\sum |u_n|$ converges.
- Conditional Convergence:** $\sum u_n$ converges, but $\sum |u_n|$ diverges.

2. Fourier Series & Half-Range Expansions

Dirichlet's Conditions: $f(x)$ has Fourier expansion if in interval: (1) single-valued & bounded, (2) finite number of points of discontinuity, (3) finite number of maxima and minima.

Fourier Expansion in $[-L, L]$:

$f(x) = (a_0 / 2) + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right]$

- $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$
- $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$
- $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$

Even vs. Odd Functions in $[-L, L]$:

- Even $[f(-x) = f(x)]$:** $b_n = 0 \implies$ Fourier Cosine Series: $a_0 = \frac{2}{L} \int_0^L f dx, a_n = \frac{2}{L} \int_0^L f \cos \frac{n\pi x}{L} dx.$

- Odd $[f(-x) = -f(x)]$:** $a_0 = 0, a_n = 0 \implies$ Fourier Sine Series: $b_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L} dx.$

Half-Range Expansions in $[0, L]$: Extended as even (Cosine) or odd (Sine).